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NEURAL MASS AND SPIKING MODELS FOR THE STUDY OF NEURAL CIRCUITS

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Abstract

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Chapter 1

Stochastic Resonance in the Honeybee Antennal Lobe

1.1 Introduction

One of the first well established findings in neuroscience with the advent of electrical recordings of neuronal activity, was the variability of neurons' responses. That is, when stimulated with the same input, neurons respond with different temporal patterns of spikes, while the overall firing rate remains roughly equal [1–3].

This trial-to-trial variability is commonly called noise. Sources of noise in neurons are many and not all variability in response is due to noise (see [4] for a review). Still, due to the physical scale of cells, a stochastic component in neuronal function is inevitable by thermal noise alone. It is then to be expected that neural systems would have evolved to tolerate or, more daringly, exploit the presence of noise.

1.1.1 Classical stochastic resonance

The idea of studying the effects of noise on the dynamics of a system first appeared parallely in the work of Benzi et al. [5] and Nicolis [6], where the term Stochastic Resonance (SR) was used to explain the periodic behavior of earth's average temperature.

A brief covering of that first SR application will be made, as it provides a practical understanding of the general phenomenon and a surprising link to neural systems.

The most common way to model temperature averages was using simple energy balance deterministic models:

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = R_{in}(T) - R_{out}(T) \tag{1.1}$$

where C is the earth's thermal capacity, and R_{in} and R_{out} represent the total incoming and outgoing solar radiation, defined as

$$\begin{aligned} R_{in} &= Q\mu \\ R_{out} &= \alpha(T)Q\mu + \varepsilon(T) \end{aligned}$$

where μ is the average incoming solar radiation.

In [7] a simplification is used: 3 steady-states are assumed to exist as $T_1 < T_2 < T_3$, those are possible earth's climates. 1.1 then becomes

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{\mu\varepsilon(T)}{1 + \beta\,(1 - T/T_1)\,(1 - T/T_2)\,(1 - T/T_3)} - \varepsilon(T) \tag{1.2}$$

To analyze the stability of the defined climates the rate of change of temperatures is taken as

$$F(T) = \frac{\mathrm{d}T}{\mathrm{d}t}\ , \tag{1.3}$$

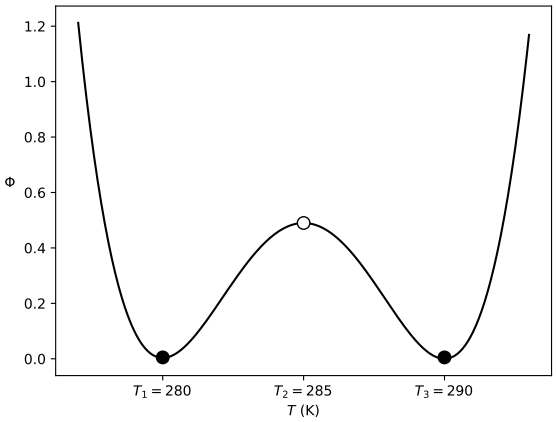


Figure 1.1: The double well potential. Filled circles are stable points.

then a pseudo-potential energy function is defined as

$$\Phi(T) = -\int F(T)\mathrm{d}T \tag{1.4}$$

that represents the "energy" needed to change the temperature across temperature values. From Φ we get

$$F(T) = -\frac{\partial \Phi}{\partial T} \tag{1.5}$$

Then intuitively the maximum of Φ will be an unstable point, while the minima will be stable. That is, where the second derivative with respect to temperature

$$\frac{\mathrm{d}F}{\mathrm{d}T} = -\frac{\partial^2 \Phi}{\partial T^2} \tag{1.6}$$

is < 0, any neighboring starting state of the earth's temperature will converge to it, while when it is > 0 they will diverge.

It can be shown that T_1, T_3 are stable points, while T_2 is unstable: a peak any starting state would "roll off" from (see A). In our context, T_1, T_3 are the only observable climates, temperatures at which the earth will stay for enough time to be detectable in the geological records. A system where 2 stable states separated by an unstable state exist is called a bistable system, and the plot of Φ over T displays the typical double well shape of the (pseudo)potential function (Figure 1.1).

With the present model, we can see that the earth's climate would depend solely (all else being equal) on the starting temperature, i.e., whether it is smaller or greater than T_2. Then it would follow $F(T)$ towards T_1 or T_3 (Figure 1.2). However, earth's temperatures do indeed show oscillations with a period of 10^5 years.

A weak cosinusoidal forcing modulating the input solar radiations can be introduced, representing the varying distance of the earth from the sun along its elliptic orbit. Then μ becomes time dependent

$$\mu(t) = 1 + \gamma \cos(\omega t) \tag{1.7}$$

where $\gamma \ll 1$ and $\omega = 2\pi/10^5$, reflecting the observed oscillation period. The derivative and pseudo-energy function become time dependent $\tilde{F} = \tilde{F}(T;t)$, $\tilde{\Phi} = \tilde{\Phi}(T;t)$.

Still, this periodicity alone does not lead to any change in climates: the system never jumps over T_2 (see fig).

If gaussian white noise, representing various mechanisms causing T fluctuations at a smaller time scale, is added to the system, yielding

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \tilde{F}(T;t) + \sigma\eta(t) \tag{1.8}$$

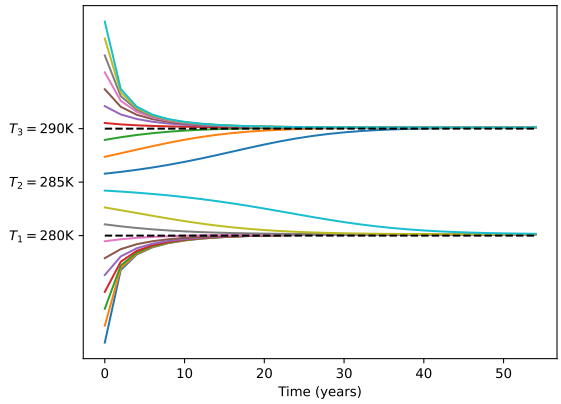


Figure 1.2: Trajectories for different starting temperatures without noise.

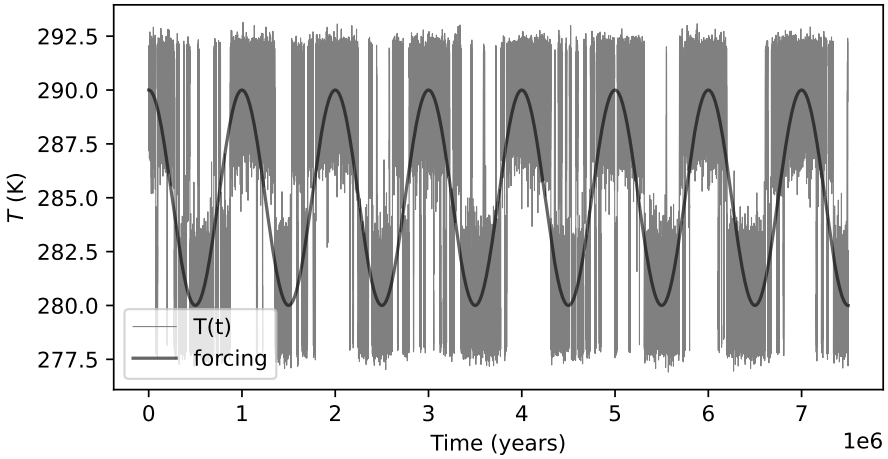


Figure 1.3: Temperature over time from numerical solution of 1.2, with $\sigma = 0.15$.

then for certain values of σ a jump between stable states becomes possible, with a frequency $\approx 2\pi/10^5$. That is, the system shows temperature oscillations at the forcing frequency, as noise allows the system to jump from one equilibrium to another (Figure 1.3). Note that oscillations arise from the joint effect of the periodic forcing and noise: noise pushes the system away from the equilibrium at every timestep, but only when the forcing asymmetrically tilts the double well potential such that the distance between the current equilibrium and the unstable one becomes small enough to allow the stochastic component to push the system over it, a climate transition (that is, a jump from T_1 to T_3 or viceversa) can occur. This can be seen as an "escape" from a stable equilibrium, requiring a jump over a (pseudo)energy potential "wall". This is a classic problem in physics, Kramer's rate [8] gives the frequency of jumps per second

$$r_K = \frac{\sqrt{V''(x_m)/m}\sqrt{|V''(x_b)/m|}}{2\pi\gamma} e^{-\frac{\Delta V}{D}} \tag{1.9}$$

where $V''(x_m)$ is the curvature around the minima, $V''(x_b)$ that at the maximum, ΔV the distance from the minimum to the maximum, and $D = \sigma^2/2$ the temperature of the system (noise intensity in our context). The inverse $\frac{1}{r_K}$ gives the residence time, the time spent at one of the minima. If our condition for SR is that it allows the system to follow the subthreshold forcing, then the optimal D is such that, given the forcing period T_ω and

the residence time as a function of D, $T_K(D) = \frac{1}{r_K(D)}$ is

$$T_K(D) = \frac{T_\omega}{2} \tag{1.10}$$

Some useful considerations can be made on SR defined on the climate model. First, it requires a system containing a cubic nonlinearity, as in 1.2, such that 2 stable states separated by a potential wall exist. Additionally, the periodic forcing must be insufficient alone to cause a crossing (subthreshold). Later it will be shown that both of these assumptions can be relaxed. Last, as 1.10 suggests, too low D will not be able to push the system over the barrier at the frequency needed, and high enough D make the potential barrier effectively nonexistent, increasing r_K and making the double well model meaningless.

Hence, given a metric of how well the system follows the forcing, there will be a peak for a certain D. In this sense, there is a resonance in the system. For classic SR models, the natural measure is the spectral power (obtained by Fourier transform of the system response) at the forcing frequency relative to the neighboring ones: Signal-to-Noise Ratio (SNR) [9].

1.1.2 Small detour on dynamical systems

It will be shown that neurons are better modeled by 2D systems, due to the coexistence and interaction of 2 main ionic currents that govern a neuron's state $\left(\mathrm{Na}^+ \text{ and } \mathrm{K}^+\right)$, that determine a depolarization-hyperpolarization feedback loop. To analyze a 2+D systems the tools of dynamical systems theory are indispensable. A brief covering of the main methods will follow, as they will be referenced in the following chapters (for a more complete treatment, see [10]).

For any 2D system defined by

$$\dot{v} = f(v, u) \tag{1.11}$$

$$\dot{u} = g(v, u) \tag{1.12}$$

the system exists in the space $S = \{v\mathbf{e}_1 + u\mathbf{e}_2 \mid v, u \in \mathbb{R}\}$, defined by a set of parameters $\mathcal{P}$, and always identified with the state vector $\mathbf{s} = (v, u)$.

Then we define the mapping $F : \mathbb{R}^2 \to \mathbb{R}^2$ such that $F : \mathbf{s} \mapsto (f(\mathbf{s}), g(\mathbf{s}))$, the standard definition of a vector field (in $\mathbb{R}^n$). The direction of $F(\mathbf{s})$ is given by the sign of $f(\mathbf{s})$ and $g(\mathbf{s})$. So the curves defined by $f(v, u) = 0$ and $g(v, u) = 0$, the v- and u-nullclines, that separate the plane in changes in the sign of $\dot{v}$ and $\dot{u}$, are of particular interest. Note that $\dot{v} = \mathrm{d}v(t)/\mathrm{d}t$ and $\dot{u} = \mathrm{d}u(t)/\mathrm{d}t$. Then the evolution of the system's state from an initial position $\mathbf{s}(t_0) = (v(t_0), u(t_0))$ up to a time t, $\mathbf{s}(t)$, is governed by the flow of the vector field, formally $\varphi^t : S \to S$. Indeed, the trajectory $\varphi^{t-t_0}(\mathbf{s}_0)$ is a curve always tangent to F.

All points of intersection of the nullclines are such that $\dot{v} = \dot{u} = 0$. Consider the state $\mathbf{s}^*$ such that $f(\mathbf{s}^*) = g(\mathbf{s}^*) = 0$, if the system remains at $\mathbf{s}^*$ for all $t \geq 0$ then $\mathbf{s}^*$ is an equilibrium. Equilibria can be stable or unstable, the stability of an equilibria can be probed by analyzing the response of the system near $\mathbf{s}^*$ to a small disturbance $\tilde{\mathbf{s}} = \mathbf{s} - \mathbf{s}^*$, such that $|\mathbf{s}^* - \mathbf{s}| \ll 1$. Note that by definition $F(\mathbf{s}^*) = \mathbf{0}$, so $\mathbf{s}^*$ is treated as a constant, and $\dot{\tilde{\mathbf{s}}} = F(\mathbf{s}^* + \tilde{\mathbf{s}})$. Then, in time domain, we are interested in $\varphi^t(\tilde{\mathbf{s}})$, for some time $t > t_0 = 0$.

However, the trajectory based on f, g usually cannot be calculated when they are nonlinear (as is the case for 1.18). Then we use the linearization of the system near the equilibria, solving

$$\dot{\tilde{\mathbf{s}}} = \mathrm{J}\tilde{\mathbf{s}} \equiv \begin{pmatrix} \dot{\tilde{v}} \\ \dot{\tilde{u}} \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{\partial f(\mathbf{s}^*)}{\partial v} & \frac{\partial f(\mathbf{s}^*)}{\partial u} \\ \frac{\partial g(\mathbf{s}^*)}{\partial v} & \frac{\partial g(\mathbf{s}^*)}{\partial u} \end{pmatrix}}_{\mathrm{J}} \begin{pmatrix} \tilde{v} \\ \tilde{u} \end{pmatrix} \tag{1.13}$$

note that in the linearization of the system quadratic and higher orders terms are omitted since they are evaluated on the small disturbance $\tilde{\mathbf{s}}$, and so are assumed to contribute little to the solution.

To solve the system above we can use the eigenvalue method, solutions are in the form $\tilde{\mathbf{s}}(t) = e^{\lambda t}\boldsymbol{\xi}$, where $\boldsymbol{\xi}$ is the eigenvector. substituting into 1.13 we get

$$\dot{\tilde{\mathbf{s}}} = \mathrm{J}\tilde{\mathbf{s}} \rightarrow \lambda e^{\lambda t}\boldsymbol{\xi} = \mathrm{J}e^{\lambda t}\boldsymbol{\xi} \rightarrow \mathrm{J}\boldsymbol{\xi} = \lambda\boldsymbol{\xi} \tag{1.14}$$

The eigenvalues of J are found solving the characteristic equation $\det(\mathrm{J} - \lambda\mathrm{I}) = 0$, for 2×2 matrices, this leads to a quadratic polynomial that is solved as

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2} \tag{1.15}$$

where $\tau = \mathrm{Tr}(\mathrm{J})$, $\Delta = \det(\mathrm{J})$, note that each eigenvalue will have its corresponding eigenvector, so our example system has $\boldsymbol{\xi}_1$, $\boldsymbol{\xi}_2$.

Finally, the eigenvectors of J form the eigenbase of the system. That is, their linear combination gives the linear approximation of $\varphi^t(\tilde{\mathbf{s}})$:

$$\dot{\tilde{\mathbf{s}}} = c_1 e^{\lambda_1 t - t_0}\boldsymbol{\xi}_1 + c_2 e^{\lambda_2 t - t_0}\boldsymbol{\xi}_2 \tag{1.16}$$

where c_1, c_2 are determined by $\tilde{\mathbf{s}}$.

Clearly, the behavior near the equilibria will be a linear combination of exponential motion along $\boldsymbol{\xi}_1$, $\boldsymbol{\xi}_2$ with direction determined by the signs of λ_1, λ_2. Each equilibria of a system can then be categorized by its eigenvalues.

The main types of equilibria relevant to this work are

- Nodes: real eigenvalues with the same sign. If $\lambda_{1,2} < 0$ the node is stable: trajectories converge to it. If $\lambda_{1,2} > 0$ unstable: trajectories diverge from it.
- Saddles: real eigenvalues with opposite sign. Thus always unstable.
- Focuses: complex-conjugate eigenvalues (when $\tau^2 - 4\Delta < 0$), trajectories spiral either inward or outward depending on stability. Sign of $\Re(\lambda_{1,2}) \neq 0$ determines the stability, while $\Im(\lambda_{1,2})$ determines the frequency of the rotation.

Knowing type and stability of equilibria, joined with knowing F, provides information on all possible behaviors of a system, given its initial state.

1.1.3 From Earth to neurons

Historically the first model used to study SR was that of 1.2, more generally expressed as a stochastic differential equation [5]

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x\left(a - bx^2\right) + \mu\gamma\cos(\omega t) + \sigma\xi(t) \tag{1.17}$$

where ξ is gaussian white noise, and $x(a - bx^2)$ cubic term, that for $a > 0$ has 3 fixed points. When defining the pseudo-potential as in 1.4 it gives the quartic polynomial that is the simplest realization of the double potential well (here ignoring forcing and noise component)

$$\begin{aligned}\Phi &= -\int -bx^3 + ax \\ &= \frac{b}{4}x^4 - \frac{a}{2}x^2 + C\end{aligned}$$

Intuitively, a neuron is a system that displays both quiescent and spiking behavior: that is, 2 stable states, separated by a "barrier" commonly called the voltage threshold. Note that the term threshold is hardly accurate: different current values can all elicit spikes in neurons depending on stimulation duration, some neurons respond to certain stimulation frequencies at currents lower than those needed to elicit a spike with a pulse-like stimulation, and others fire in response to hyperpolarizing currents (known as a rebound spike) [11, Chapter 7].

Unsurprisingly, a neuron can be modeled in the same form of 1.17 to show SR [12]. Interestingly, the same holds for more accurate neuron models.

1.1.4 The FitzHugh-Nagumo model

The FitzHugh-Nagumo (FHN) model [13, 14] is a 2D reduction of the Hodgkin-Huxley (HH) biophysical model [15], its attractiveness stems from being able to recreate the spiking behavior of neurons without the 4 differential equations that regulate the HH neuron, allowing for easier analysis of the model using the theory of dynamical systems.

The FHN model is defined by a fast and a slow variable

$$\frac{\mathrm{d}v}{\mathrm{d}t} = v\,(a - v)\,(v - 1) - w + I \tag{1.18}$$

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \varepsilon(bv - cw) \tag{1.19}$$

with v representing the (fast) membrane voltage and w the (slow, with $\varepsilon \ll 1$) feedback recovery mechanisms of the membrane (e.g., opening of K^+ channels). The main difference of this model with 1.2 is the second equation for w that allows for an additional behavior that cannot be seen in the climate model. However, a cubic term for $\dot{v}$ is present and one may wonder whether the resulting double well potential is a requirement for SR. It will be shown that it is not, and that it would describe an unrealistic regime for neural function under SR.

To see how the 2 models are related, both can be analyzed with the methods of 1.1.2. In the case of $\varepsilon \to 0$, we can treat w as a constant, then it is clear that the FHN model's phase portrait is topologically equivalent to 1.17 (see fig) for the right $\mathcal{P}$ (see B).

However, the FHN 1d reduction does not have a spiking mechanism, and the resulting neuron model would only show 2 resting states at different membrane voltages values v_1, v_3, exactly as in 1.2. Thinking about biological neurons, we would be describing a neuron with a single ionic current with a depolarizing effect, so voltage would only grow, with no recovery feedback. Clearly the 1d model is insufficient. Inspecting the full FHN we note that it is the slow w, entering as a subtractive term in v, that causes additional repolarization of the membrane, mapping to our biological understanding of the spiking mechanism.

Then the nonlinearity we are interested in is whichever is born from the interaction of the 2 opposing forces.

1.1.5 Neuron types and inputs

Different $\mathcal{P}$ construct different fields F, resulting in different equilibria, both number and type, and bifurcations; all leading to different behaviors. Thus the FHN can model different types of neurons, the main types are known as integrators (type-1) and resonators (type-2). The 2 types have different computational properties, but both can rightly be used for primary and secondary sensory neurons (which are of interest in this chapter), depending on the type of stimulus they are supposed to receive and process.

Stimuli can be modeled as instantaneous pulse-like or smooth ramp-up/down inputs. It is useful to clarify what is intended for instantaneous input (or shift in v), as it is also

the formalization of a spike event in more simplified models, that will be used later. The common mathematical object used is Dirac's δ

$$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases} \tag{1.20}$$

such that

$$\int_{-\infty}^{\infty} \delta(x)dx = 1 \tag{1.21}$$

Then, defining the stimulation time as t_0, 1.18 becomes

$$\frac{\mathrm{d}v(t)}{\mathrm{d}t} = v\,(a - v)\,(v - 1) - w + I(t) \tag{1.22}$$

$$\frac{\mathrm{d}w(t)}{\mathrm{d}t} = \varepsilon(bv - cw) \tag{1.23}$$

where $I(t) = Q\delta(t - t_0)$, and Q is the voltage jump (more correctly Q/C, but the membrane's capacitance is assumed $C = 1$). Essentially $I(t)$ at t_0 shifts the whole system state vector to a new position in the state space $\tilde{\mathbf{s}} = (v + Q,\, w + 0)$, acting as a macro-perturbation from equilibrium.

It is to note that the δ-like input is one limited way to model the stimulation that a neuron may be subjected to. Often a neuron may receive multiple inputs from pre-synaptic elements that can vary in intensity, depending on many biological details. Thus, another way to model stimulation is to allow $I(t)$ to vary continuously with time, mimicking the smoothing summation of incoming inputs at different timings.

From the perspective of dynamical systems, the effect is not of a perturbation on $\mathbf{s}$. Rather, when the changes in I are slow enough, the geometry of the phase portrait itself changes, as w changes accordingly. Effectively I is to be thought of as another parameter in $\mathcal{P}$ that we choose to vary.

As seen in 1.1.2, this is the setting for a codimension-1 bifurcation. Then, unlike δ inputs, smooth ones result in a change in the geometry of the phase portrait of the system, changing its overall behavior.

In 1.1.1 we saw that SR happens for an input $\mu(t)$ that was unable by itself to lift the system over the pseudopotential "wall" (intuitively no better tracking of $\mu(t)$ could be made otherwise). It is clear that for neurons the "wall" is the difference between the spiking and non-spiking state. Therefore, it is now necessary to clarify the meaning of threshold.

1.1.6 Neural nonlinearity and threshold phenomena

The usual working definition of a threshold is a membrane voltage value v_ϑ that when reached is followed by a large excursion in v, while no excursion is present for $v < v_\vartheta$. The discontinuity in the trajectories of v near v_ϑ is our wall to be crossed. However, looking at the vector fields depicted in the phase portraits, it is unclear whether there really is such a threshold value.

Indeed, consider the FHN in its simplest configuration: having a unique globally attracting stable equilibrium, meaning that any excitation of the system will eventually converge to it. This mirrors our understanding of a neuron with a preferred resting state that after stimulation relaxes back to it.

The FHN system so defined does not have any discontinuity in its vector flow, no all-or-none response can arise theoretically (see C for an argument). In fig [fig monstable] such a system is depicted. The phase portrait is divided by the w/v-nullclines in 4 regions, depending on the direction of $(\dot{v}, \dot{w})$, a δ-like stimulus may push the system state vector $\mathbf{s}$

in either the $(\dot{v} < 0, \dot{w} > 0)$ or the $(\dot{v} > 0, \dot{w} > 0)$ regions. In the former case $\mathbf{s}$ will quickly return to $\mathbf{s}_0$, while in the latter it will go quickly to the right of the unstable middle branch of the v-nullclines and cross it after its right knee, then coming back to $\mathbf{s}_0$ as $\dot{v} < 0$.

However, for the right stimulation, the system may also travel along the middle branch of v-nullcline up to the right knee before jumping to a stable branch. in fact, there exist a set of those intermediate trajectories (canards), that trace along the slow manifold: the v-nullcline where $\dot{v} = 0$, and the dynamics are governed by the slow w variable.

Variable amplitude action potential are also present in biological neurons [16], but the overwhelming majority of observations points to single threshold values and all-or-none responses. This is due to the thinness of the set of canards: given the instability of the middle branch few trajectories follow it for enough time to produce the intermediate response. Indeed, The size of the interval is of order $O(e^{-c/\varepsilon})$, where c is a positive constant and $\varepsilon \ll 1$ is the separation between the fast and slow time scales [17]. Then in the physical world channel noise alone would make it impossible to remain on the slow manifold, that can be regarded in realistic conditions as a "quasi-threshold" or a threshold manifold [18].

Monostable models are usually near an Hopf bifurcation, hence resonators. Integrators are instead modeled by systems near saddle-node bifurcations, then at least a node and a saddle must exist in the state space. Assuming a δ input that shifts the state vector $\mathbf{s}$ from the stable node (again, the resting state), if Q moves $\mathbf{s}$ to the left of the stable branch of the saddle point, $\mathbf{s}$ trajectory is first dominated by the negative eigenvalue, then by the positive, being repelled back to the node. That is, no large excursion happens. When Q is large enough to move $\mathbf{s}$ to the right of the stable branch, the system follows again the eigenvector with the negative eigenvalue, but is then repelled to the right, resulting in a large excursion in v (see Fig [saddle-node fig]). Therefore in integrators there exist a real discontinuity: the stable branch of the saddle, called the separatrix, represents a set of thresholds [19].

For non δ stimulation the threshold may be viewed as the critical value of the codim-1 parameter (i.e., the input voltage), whenever it is crossed the system undergoes a bifurcation that changes its behavior. Usually in FHN and similar models this leads to a continued spiking state, with frequency dependent on the input for saddle-node and a roughly stable intrinsic frequency in hopf bifurcations.

Various models and (quasi)threshold phenomena have been described, returning to the arguments in 1.1.1, it is clear that nonlinear elements in neurons create different types of nonlinearity in responses. However, once biological parameter ranges are taken into consideration, we may define the general neuronal nonlinearity as the presence of a threshold response.

Note that the different dynamical properties of the models have an important role in the neurocomputational characteristics of neurons, but, for the purpose of studying SR for recovery of an external input, that simple concept is all we need. Indeed, SR is found in different dynamical configurations of the FHN [20–22].

1.1.7 Information in threshold elements

Considering a threshold as the nonlinearity needed for SR to arise simplifies the analysis. We start with the simplest possible threshold unit, with response y(x(t)), function of time dependent input x(t) such that

$$y(x(t)) = \begin{cases} 1 & \text{if } x(t) \geq \theta \\ 0 & \text{if } x(t) < \theta \end{cases} \tag{1.24}$$

if $\max_{t\in\mathbb{R}} x(t) < \theta$ or $\min_{t\in\mathbb{R}} x(t) \geq \theta$ then intuitively we can see that no information on the stimulus can be transmitted by $y(x(t))$.

We can find an helpful formalization of this notion in information theory, where the analysis of the element in 1.24 is a classic result, see [23].

Considering any random variable X with probability mass function $p(x)$ and support $\mathcal{X}$, the uncertainty we have by observing samplings of X is called entropy, defined as

$$H(X) = -\sum_{x\in\mathcal{X}} p(x)\log p(x) \tag{1.25}$$

note that herein we take $\log = \log_2$, expressing the quantity in bits, and we take $0\log 0 = 0$.

Assuming we have another random variable Y, related in some way to X, the remaining average uncertainty on X given Y is the conditional entropy

$$\begin{aligned} H(X|Y) &= \sum_{y\in Y} p(y)H(X|Y=y) && (1.26)\\ &= -\sum_{y\in\mathcal{Y}} p(y)\sum_{x\in\mathcal{X}} p(x|y)\log p(x|y) && (1.27)\\ &= -\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\log p(x|y) && (1.28)\end{aligned}$$

Especially useful for the treatment of SR is the Mutual Information (MI) between two random variables. Considering discrete X, Y the MI between them can be understood as the reduction in uncertainty on X, knowing Y, or vice versa, due to the symmetric relationship of the marginal and joint distributions.

$$\begin{aligned} I(X;Y) &= H(X) - H(X|Y) && (1.29)\\ &= -\sum_{x\in\mathcal{X}} p(x)\log p(x) - \left(-\sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y)\log p(x|y)\right) && (1.30)\end{aligned}$$

A consequence is that the MI between any variable X and itself is equal to $H(X)$, as $H(X|X) = 0$, so we can see entropy as a sort of self-information.

Then with 1.24 we can construct the probability mass distribution of our outputs Y. For a strictly supra or sub -threshold stimulus we trivially see that taking $\mathcal{Y} = \{0, 1\}$ we get a concentrated probability mass either at $P(Y = 1) = 1$ or $P(Y = 0) = 1$, respectively. Then the self-information of our output, regardless of the input, is always 0, as

$$\begin{aligned} H(Y) &= -\left(P(0)\log P(0) + P(1)\log P(1)\right) && (1.31)\\ &= -\left(1\log 1 + 0\log 0\right) \text{ or } -\left(0\log 0 + 1\log 1\right) && (1.32)\end{aligned}$$

Instead, if we assume the distribution of the signal X to be gaussian with $\mathbb{E}[X] = \theta$, we know that $P(X \geq \theta) = 0.5$, then $P(X < \theta) = 1 - P(X \geq \theta)$, that gives $P(Y = 1) = 0.5$, then $P(Y = 0) = 1 - P(Y = 1)$. So finding the self-information of the output

$$H(Y) = -\left(\frac{1}{2}\log\frac{1}{2} + \frac{1}{2}\log\frac{1}{2}\right) = 1 \tag{1.33}$$

note that 1 bit is hence the upper bound of the information an element of the form 1.24 (called a binary channel) can transmit about any stimulus.

This holds in a deterministic system, as $H(Y|X) = 0$ gives $I(Y;X) = H(Y) - 0$. However, if noise is added to the element, meaning that an output that would be 1 can change to 0 and vice versa with probability p', then $H(Y|X)$ becomes nonzero, depending entirely on the noise entropy

$$H(p') = -\left(p'\log p' + (1-p')\log(1-p')\right). \tag{1.34}$$

To show this, it is convenient to map the continuous input X to the discrete X', via the indicator function

$$X' = \mathbb{I}(X \geq \theta) \tag{1.35}$$

resulting in X' having a discrete support $\mathcal{X}' = \{0, 1\}$, with the probabilities $P(X' = 1) = P(X \geq \theta) = P(X' = 0) = P(X < \theta) = 1/2$. The conditional entropy of the output given the input becomes

$$\begin{aligned}
H(Y|X') &= \sum_{x' \in \mathcal{X}'} p(x') H(Y|X' = x') \\
&= -\sum_{x' \in \mathcal{X}'} p(x') \left(P(Y = 1|x') \log P(Y = 1|x') + P(Y = 0|x') \log P(Y = 0|x') \right) \\
&= -\left[P(X' = 1) \left(P(Y = 1|1) \log P(Y = 1|1) + P(Y = 0|1) \log P(Y = 0|1) \right) \right. \\
&\quad \left. P(X' = 0) \left(P(Y = 0|0) \log + P(Y = 0|0) + P(Y = 1|0) \log P(Y = 1|0) \right) \right] \\
&= -\frac{1}{2} \left[2 \left((1 - p') \log(1 - p') + p' \log p' \right) \right] \\
&= -\frac{1}{2} \left[2 \left(-H(p') \right) \right] = H(p')
\end{aligned}$$

resulting in the MI

$$I(Y; X') = H(Y) - H(p') \tag{1.36}$$

Thus noise can only lower the amount of information in a binary channel, when the stimulus is optimal.

1.1.8 SR in networks of threshold elements

The simplicity of the single element allows to explore an ensemble of threshold elements. Considering N elements in the form of 1.24, the output of the network can be defined by a simple sum

$$y_N(x(t)) = \sum_{n=1}^{N} y_n(x(t)) \tag{1.37}$$

In such a network Stocks [24, 25], showed that SR can occur. Specifically, the phenomenon was called Suprathreshold Stochastic resonance (SSR), since the MI between input and output was increased at the optimal input distribution.

Independent noise η_i of variance σ^2 is added to each element in 1.24, so that the individual output for the $i = 1, \ldots, N$ elements becomes

$$y_i(x_i(t)) = \begin{cases} 1 & \text{if } x_i(t) + \eta_i(t) \geq \theta_i \\ 0 & \text{otherwise} \end{cases} \tag{1.38}$$

Note that for future use it is convenient to take standardized (unit variance) noise ξ, thus expressing the noise intensity directly by σ as its sd: $\eta_i = \sigma \xi_i$.

With $\sigma = 0$, all elements behave the same, the system collapses to 1.24 and no information gain can happen. However, nonzero noise added to the element is mathematically equivalent to independent variations of each θ_i, thus the number of activated elements at each t can be different than N or 0. That is, the response of the single element with nonzero noise is not a step function of the input. Rather, for each realization of $X = x$, it is influenced by the probability that noise changes the deterministic output: in the case $x \geq \theta$ the probability of the element being active

$$P(1|x) = \int_{\theta - x}^{\infty} p_\eta(\eta) d\eta \tag{1.39}$$

and for $x < \theta$ the probability of it not being active

$$P(0|x) = \int_{-\infty}^{\theta - x} p_\eta(\eta) d\eta \tag{1.40}$$

If we assume symmetric distribution for both η and X, then both are equal, and depend only on the distance from threshold $d = |x - \theta|$.

Considering the whole network, the active probability can be used to obtain the probability of a given count n given a certain realization of X.

$$P(n|x) = P(1|x)^n (1 - P(1|x))^{N-n} \tag{1.41}$$

since individual crossings at each d are dependent only on the individual independent noise, we care about the count, so we consider all possible combination of the N that give a count n: the binomial

$$P(n|x) = \binom{N}{n} P(1|x)^n (1 - P(1|x))^{N-n}. \tag{1.42}$$

Averaging across X we get the overall probability of having an individual n count response, and of all possible combination giving n counts, respectively

$$\mathbb{E}[\mathbf{y}|X] = \int_{-\infty}^{\infty} dx p_X(x) P(1|x)^n (1 - P(1|x))^{N-n} = P'(n) \tag{1.43}$$

$$\mathbb{E}[Y_n|X] = \int_{-\infty}^{\infty} dx p_X(x) \binom{N}{n} P(1|x)^n (1 - P(1|x))^{N-n} = P(n) \tag{1.44}$$

Substituting into 1.29 (with X continuous) we get

$$I(Y;X) \tag{1.45}$$

$$= -\sum_{n=0}^{N} P(n) \log P'(n) - \left\{ -N \int_{-\infty}^{\infty} dx p_X(x) \left[P(1|x) \log P(1|x) + P(0|x) \log P(0|x) \right] \right\} \tag{1.46}$$

Assuming that $p_\eta(\eta)$ and $p_X(x)$ are equal except for their first moment, and that $\mathbb{E}[X] = \theta$, we get the approximated closed form solution (see D)

$$I(Y;X) = \frac{1}{2} \log(N+1) - \frac{1}{2 \log_e 2} + O(1/N) \tag{1.47}$$

That shows how I grows with N. Additionally, since the case $\sigma = 0$ yields $P(n) = 0$ for all counts $n \neq N, 0$, as seen previously the network's I is capped at 1 bit. Then it is evident that for high enough N, for a given $\sigma \neq 0$, we will have $I > 1$ bit. For $\sigma \to \infty$, the individual outputs are completely dependent on noise, hence $H(Y) = H(Y|X)$ and $I(Y;X) = 0$.

In this sense we are sure that there must exist an optimal value of σ for which $I(Y|X)$ is maximized, that will depend on both noise and stimulus distributions.

Specifically, noise influences both the summation and integral of the RHS of 1.45, without specifying $p_\eta(\eta), p_X(x)$, and the optimal σ value will be such that

Another view on the mechanism of SR in the network is that noise effectively makes the step-like response function of the single element a graded function of the stimulus, within a certain window (see 1.39,1.40). When considering the output of a single element, which is still binary, the linearized response cannot be sampled accurately, but when considering the sum of $N > 1$ elements, it is possible to get a discrete approximation.

1.1.9 Real neurons and MI

The appeal of the summing network approach to the study of SR in neuroscience comes from the fact that it may serve as a starting approximation of a network of neurons. Indeed, if mapping the 1 output of the threshold element to the spike event, it would seem that it is a good approximation: neurons integrate incoming signals and spike when $v_m \geq v_\theta$, and a possible readout of their activity is the spike count.

However, the information gain on the negative d between threshold and stimulus happens only because in the noiseless element the only output is 1 at any time t (and as already seen, the noiseless network collapses to the single element output). While also a neuron has a single output state, in neural function its output is usually meaningless if not integrated by a receiving neuron. A spike is received as a graded change in membrane voltage, that decays exponentially in time and space. Given that in most neurons a received spike is not enough to make $v_m \geq v_\theta$, then in the simplified case of two neurons in an input-output relationship, what allows for transmission of information is the rate of the spiking events from the input, that will be translated in an output rate.

The discrete spike events are thus mapped to a continuous firing rate defined within an arbitrary window, chosen such that it is relevant to the biological integration time constant. Neurons commonly found in sensory systems are integrators, as seen in 1.1.6 the underlying dynamics of integrators are such that they respond approximately linearly to the total incoming input, when above rheobase. As will be shown, this makes the definition of SSR for threshold elements misleading in more biologically accurate models.

Still, the framework developed for simple threshold elements is useful as it gives a precise measure of "performance" that does not rely on periodicity of the signal as the SNR. Indeed, removing assumption on the type of signal, information theoretic measures allow for a more general characterization of the phenomenon, more so when studying biological neural networks, where the possible input distributions cannot be easily assumed.

1.1.10 SR in biological models

In the climate model noise allowed the periodic changes of total incoming solar radiation that were too weak to move the earth systems temperatures alone, to influence it.

Having generalized the needed nonlinearity to the notion of threshold, we see the possible role of SR in neural function: to allow for the detection of a signal too weak to be detected otherwise. Clearly, this problem is primarily present in sensory systems. Indeed, SR in sensory tasks was found in various animal models [26–30]. However, experimental approaches do not manipulate internal noise in the animal neurons, nor can they easily test a range of $p_X(x), p_\eta(\eta)$. Though it can be argued that for the single neuron case adding noise to the stimulus is mathematically equal to noise in the neuron, this and the fact that it is difficult to precisely record a large network of neurons limit the experimental studies to single neuron or behavioral effects. Conversely, most theoretical and simulation studies focus on simple networks that are not directly comparable to biological systems, especially with many details of SR depending on parameter choices that are hard to relate to their biological counterparts.

A middle ground can be found in insect models: their relative simplicity allows for precise mapping of sensory networks, and experimental manipulation and neuronal recording of large population level activity can be more easily performed, see . At the same time, once the network is isolated, it can be analyzed within a precise theoretical framework, and simulated where the purely theoretical approach would be intractable.

Here, a model of the Honeybee Antennal Lobe (AL), primary center for olfaction, is studied for its suitability for SR, from the single neuron to the whole network level. An advantage of studying olfactory networks is the remarkable similarity in the sensory

architecture among mammals and insect [31–34]. This allows for useful generalization, while also having a biological reference for parameters.

1.2 Methods

1.2.1 AL network

Appendix A

Stability of Equilibria

In Benzi et al. [7]

Appendix B

Phase portraits of climate and FHN models

Appendix C

No threshold in monostable FHN

Consider the FHN system in a configuration having a unique globally attracting stable equilibrium $\mathbf{s}_0 = (v_0, w_0)$.

Take $\dot{\mathbf{s}} = f(\mathbf{s})$ as

$$f(v,w) = \left(v - \frac{v^3}{3} - w - I, \varepsilon(v + a - bw)\right) \tag{C.1}$$

then $f : \mathbb{R}^2 \to \mathbb{R}^2$ is smooth on R^2. We define a stimulation function continuous on a compact set of inputs values $Q = [0, q_{max}] \subset \mathbb{R}$, modeling an instantaneous stimulation

$$\gamma : Q \to \mathbb{R}^2,\ q \mapsto (v_0 + q, w_0) \tag{C.2}$$

then $A = \gamma(Q) \subset \mathbb{R}^2$ is the compact set of starting states after stimulation. We are interested in the properties of the flow originating from A for all $t \geq 0$.

By definition of a stable equilibrium, for any neighborhood $U \supset \mathbf{x}_0$ there exist a $V \supset \mathbf{x}_0$ such that $\varphi^t(\mathbf{x}) \in U$ for all $\mathbf{x} \in V$ and $t > 0$. That is, once entered in V, the orbit never leaves U (Lyupanov stability). Because of global attraction of $\mathbf{s}_0$, it is guaranteed that there exist a time for which $\varphi^t(\mathbf{a})$ is inside any $W \supset \mathbf{s}_0$. Then we take $\mathrm{T}^e(B(\mathbf{a}))$ as the time in which all $\varphi^t(B(\mathbf{a})) \in V$ for a sufficiently small neighborhood $B(\mathbf{a}) \supset \mathbf{a}$. Since A is compact, there are always finitely many such $B(\mathbf{a}) \in A$.

Hence we can define

$$\mathrm{T}^* = \max\left\{\mathrm{T}^e(B(\mathbf{a}_1)), \ldots, \mathrm{T}^e(B(\mathbf{a}_n))\right\} \tag{C.3}$$

as the time at which all orbits starting at A are guaranteed to be in U, it follows that $\{\varphi^t(A)\}_{t\in[0,\mathrm{T}^*]}$ is a bounded set of continuous trajectories.

For Lyupanov stability, once entered V all orbit stay in U for all time, and for asymptotic stability of $\mathbf{s}_0$ all $\varphi^t(\mathbf{x}) \to \mathbf{s}_0$ as $t \to \infty$. So $\{\varphi^t(A)\}_{t>\mathrm{T}^*}$ is bounded.

Now we consider the complete flow from A as the closed set

$$C = \{\varphi^t(A)\}_{t\in[0,\mathrm{T}^*]} \cup \{\varphi^t(A)\}_{t>\mathrm{T}^*} \tag{C.4}$$

since both elements are bounded, C is a closed and bounded set in $\mathbb{R}^2$, so compact.

Therefore smooth f is Lipstichz continuous on C, that is, there exist a uniform L such that

$$|f(x) - f(y)| = L\,|x - y| \tag{C.5}$$

for all $x, y \in C$, and Gronwall's inequality applies

$$|\varphi^t(x) - \varphi^t(y)| = e^{Lt}\,|x - y| \tag{C.6}$$

for all $x, y \in C$. Thus for any closed interval in A, the resulting flow is always continuous.

To see the consequences of this: let $v(\mathbf{s})$ be the function giving the v value of the system state vector at any t. Then on any arbitrary closed interval $\mathcal{I} = [0, T]$ we take $\mathbf{s}_t = \varphi^t(\mathbf{a})$ for $\mathbf{a} \in A$. We can define the maximum v for a given trajectory in $\mathcal{I}$

$$V_m(\mathbf{a}) = \max_{t \in \mathcal{I}} \{v(\varphi^t(\mathbf{a}))\} \tag{C.7}$$

Considering all possible trajectories in C, C.6 applies:

$$|V_m(\mathbf{a}_1) - V_m(\mathbf{a}_2)| = e^{Lt} |\mathbf{a}_1 - \mathbf{a}_2| \tag{C.8}$$

for all $\mathbf{a}_1$, $\mathbf{a}_2 \in A$, in the arbitrary interval $\mathcal{I}$.

That is, the maximum voltage of the spike upstroke is always continuous for all stimulation values: no discontinuity in the action potential "peak" exists.

Note that as $t \to \infty$ the exponential could indeed lead to a great distance from v values. However, the maximum of a spike always happens in a finite interval after the stimulation, or it does not.

Appendix D

MI in the parallel threshold network

To arrive at the large N MI derivation in [25] we consider the output

$$Y(t) = \sum_{n=1}^{N} y_n(x(t)) \tag{D.1}$$

where the single element output is given by

$$y_n(x(t)) = \begin{cases} 1, & \text{if } x(t) + \eta_n(t) \geq \theta \\ 0, & \text{otherwise} \end{cases} \tag{D.2}$$

assuming that each $y_n(x(t))$ depend on stationary processes, we can ignore the time dependence and model the input as the random variable $X \sim (\mu_x, \sigma_x^2)$ and the independent identically distributed element noise as $\eta_n \sim (\mu_\eta, \sigma_\eta^2)$.

The formula for the MI between the continuous X and the discrete response count Y is

$$I(Y;X) = H(Y) - H(Y|X) \tag{D.3}$$

$$= -\sum_{n=0}^{N} p_Y(n) \log p_Y(n) - \left(-\int_{-\infty}^{\infty} p_X(x) dx \sum_{n=0}^{N} P(n|x) \log P(n|x) \right) \tag{D.4}$$

where the probability of each count n given a realization of x is given by the binomial

$$P(n|x) = \binom{N}{n} P(1|x)^n (1 - P(1|x))^{N-n} \tag{D.5}$$

with the otherwise deterministic output being a probability dependent on both noise and the distance of the signal from θ

$$P(1|x) = \int_{\theta - x}^{\infty} p_\eta(\eta) d\eta \tag{D.6}$$

that is, the probability that $\eta + x \geq \theta$.

Substituting D.5 the negative of conditional entropy given x expands to

$$\sum_{n=0}^{N} P(n|x) \log P(n|x) = \sum_{n=0}^{N} P(n|x) \log \left(\binom{N}{n} P(1|x)^n (1 - P(1|x))^{N-n} \right) \tag{D.7}$$

$$= \sum_{n=0}^{N} P(n|x) \log \binom{N}{n} + \log P(1|x) \sum_{n=0}^{N} n P(n|x) + \log(1 - P(n|x)) \sum_{n=0}^{N} (N-n) P(n|x) \tag{D.8}$$

$$\tag{D.9}$$

via normalization and the mean of binomial

$$= \sum_{n=0}^{N} P(n|x) \log \binom{N}{n} + NP(1|x) \log P(1|x) + N - NP(1|x) \log(1 - P(1|x)) \tag{D.10}$$

$$= \sum_{n=0}^{N} P(n|x) \log \binom{N}{n} + N \left[P(1|x) \log P(1|x) + (1 - P(1|x)) \log(1 - P(1|x)) \right] \tag{D.11}$$

then $H(Y|X = x)$ is

$$H(\mathbf{y}|X = x) - \sum_{n=0}^{N} P(n|x) \log \binom{N}{n} \tag{D.12}$$

as the second term of D.10 is the entropy of a vector output $\mathbf{y} = [y_1(x), y_2(x), ..., y_N(x)]$. This states that the conditional entropy of the count is the conditional of $\mathbf{y}$, subtracting the self-information gained by knowing the specific vector among the $\binom{N}{n}$ possible combinations generating the count n.

This matches the intuition that generally the self-information of a specific combination of outputs is higher than that of the count. However, in our case $P(y_n = 1|x)$ is dependent only on noise, given that all elements share the same θ, then we will see that MI is equal between the vector and count outputs.

Averaging over X, with $(1 - P(1|x)) = P(0|x)$

$$- \int_{-\infty}^{\infty} dx p_X(x) \left\{ \sum_{n=0}^{N} P(n|x) \log \binom{N}{n} + N \left[P(1|x) \log P(1|x) + P(0|x) \log P(0|x) \right] \right\}$$

$$= - \sum_{n=0}^{N} \log \binom{N}{n} \int_{-\infty}^{\infty} dx p_X(x) P(n|x) - N \int_{-\infty}^{\infty} dx p_X(x) \left[P(1|x) \log P(1|x) + P(0|x) \log P(0|x) \right]$$

$$= - \sum_{n=0}^{N} p_Y(n) \log \binom{N}{n} - N \int_{-\infty}^{\infty} dx p_X(x) \left[P(1|x) \log P(1|x) + P(0|x) \log P(0|x) \right].$$

Then the MI is

$$I(Y; X) \tag{D.13}$$

$$= - \sum_{n=0}^{N} p_Y(n) \log p_Y(n) - \left\{ - \sum_{n=0}^{N} p_Y(n) \log \binom{N}{n} - N \int_{-\infty}^{\infty} dx p_X(x) \left[P(1|x) \log P(1|x) + P(0|x) \log P(0|x) \right] \right. \tag{D.14}$$

$$\tag{D.15}$$

doing the first sum

$$- \sum_{n=0}^{N} p_Y(n) \log p_Y(n) + \sum_{n=0}^{N} p_Y(n) \log \binom{N}{n}$$

$$= - \sum_{n=0}^{N} p_Y(n) \left(\log p_Y(n) - \log \binom{N}{n} \right)$$

$$= - \sum_{n=0}^{N} p_Y(n) \left(\log \int_{-\infty}^{\infty} dx p_X(x) \binom{N}{n} P(1|x)^n P(0|x)^{N-n} - \log \binom{N}{n} \right)$$

$$= - \sum_{n=0}^{N} p_Y(n) \left(\log \int_{-\infty}^{\infty} dx p_X(x) P(1|x)^n P(0|x)^{N-n} + \log \binom{N}{n} - \log \binom{N}{n} \right)$$

$$= - \sum_{n=0}^{N} p_Y(n) \log p'(n)$$

where

$$p'(n) = \int_{-\infty}^{\infty} dx p_X(x) P(1|x)^n P(0|x)^{N-n} \tag{D.16}$$

is the

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